

1. Hot water at 98°C flows through a steel pipe (thermal conductivity 54 W/mK), with ID 5.25 cm , OD 6.03 cm . The inside heat transfer coefficient is $1960\text{ W/m}^2\text{K}$. The tube is exposed to atmospheric air at 20°C , the outside heat transfer coefficient is $8\text{ W/m}^2\text{K}$. Calculate the overall heat transfer coefficient based on the inside surface area and outside surface area. Consider an inside fouling factor of $0.0002\text{ m}^2\text{K/W}$ and an outside fouling factor of $0.0005\text{ m}^2\text{K/W}$.
2. A heat exchanger is used to heat water at a rate of 0.8 kg/s from 30°C to 80°C , with hot oil entering 120°C and leaving at 85°C . The overall heat transfer coefficient based on the outside surface area shall be assumed as $125\text{ W/m}^2\text{K}$. Calculate the heat exchanger area required considering (a) The heat exchanger is of parallel flow arranged (b) The heat exchanger is of counter flow arranged (c) The heat exchanger is of shell and tube arranged, with two shell pass and four tube passes. (water on shell side).
3. An oil cooler for a large diesel engine is to cool engine oil from 60°C to 45°C using sea water at an inlet temperature of 20°C with a temperature rise of 15°C . The design heat load is 140 kW . The overall heat transfer coefficient based on the outside diameter shall be assumed as $70\text{ W/m}^2\text{K}$. Calculate the heat transfer surface area for single pass counter flow and parallel flow arrangements.
4. A counter flow heat exchanger of heat transfer area 12.5 m^2 is to cool oil ($c_p = 2000\text{ J/kgK}$) with water ($c_p = 4170\text{ J/kgK}$). The oil enters at 100°C and flow rate 2 kg/s , while water enters at 20°C and 0.48 kg/s . The overall heat transfer coefficient is $400\text{ W/m}^2\text{K}$. Calculate the exit temperatures and heat transfer rate.

5. A two shell pass four tube pass heat exchanger with area 26m^2 is to cool oil at 1.5 kg/s , ($c_p = 2000\text{ J/kgK}$) with water ($c_p = 4170\text{ J/kgK}$). The oil enters at 90°C and , while water enters at 19°C and 1.0 kg/s . The overall heat transfer coefficient is $250\text{ W/m}^2\text{K}$. Calculate the exit temperatures and heat transfer rate. If the Oil flow rate is reduced to 75% of the initial flow rate, predict the exit temperatures with the same water flow rate.